

CO3 AT2

SCENARIO BASED QUESTIONS

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Question 1

Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.

SOL: Functional Dependencies (FDs)

- OrderID $\rightarrow$ CustID (an order belongs to one customer).
- CustID $\rightarrow$ CustName (a customer ID determines the customer name).
- ProdID $\rightarrow$ ProdName, Price (a product ID determines its name and price).
- (OrderID, ProdID) $\rightarrow$ Qty (the quantity of a particular product in a particular order is determined by the order and product).

Candidate Key

The candidate key is (OrderID, ProdID), because OrderID identifies the customer and ProdID identifies the product, while their combination identifies one order line and its quantity.

Problems in the Original Relation

- Update anomaly: changing a customer name requires updating multiple order records.
- Insertion anomaly: a new customer or product cannot be stored easily if there is no order.
- Deletion anomaly: deleting the only order of a customer may also remove the customer's information.

Normalization

1NF: All attributes contain atomic values, so the relation can be considered in 1NF.

2NF: Remove partial dependencies from the composite key (OrderID, ProdID). CustID depends only on OrderID, while ProdName and Price depend only on ProdID.

Relation	Attributes	Primary Key
Customer	CustID, CustName	CustID
Product	ProdID, ProdName, Price	ProdID

Orders	OrderID, CustID	OrderID
OrderItem	OrderID, ProdID, Qty	(OrderID, ProdID)

Final 3NF schema: Customer(CustID, CustName), Product(ProdID, ProdName, Price), Orders(OrderID, CustID), and OrderItem(OrderID, ProdID, Qty).

Question 2

A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.

SOL : Functional Dependencies

- $PID \rightarrow PName, DocID, WardNo$ (a patient has one name, assigned doctor, and ward).
- $DocID \rightarrow DocName, Specialization$ (a doctor ID determines doctor details).
- $WardNo \rightarrow WardName$ (a ward number determines the ward name).

Candidate Key

PID is the candidate key because PID determines all other attributes through the given dependencies.

Anomalies

- Update anomaly: a doctor's specialization may have to be changed in many patient rows.
- Insertion anomaly: a doctor or ward cannot be recorded independently without a patient record.
- Deletion anomaly: deleting the last patient assigned to a doctor or ward may remove the only stored details of that doctor or ward.

BCNF Analysis

BCNF requires that for every non-trivial FD $X \rightarrow Y$, X must be a superkey. DocID and WardNo are not superkeys of the original relation, so $DocID \rightarrow DocName, Specialization$ and $WardNo \rightarrow WardName$ violate BCNF.

BCNF Decomposition

Relation	Attributes	Primary Key
Patient	PID, PName, DocID, WardNo	PID
Doctor	DocID, DocName, Specialization	DocID

Ward	WardNo, WardName	WardNo
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In Patient, PID is the determinant and is the key. In Doctor, DocID is the key. In Ward, WardNo is the key. Therefore, all resulting relations satisfy BCNF.

Question 3

Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.

SOL : Functional Dependencies

- $SID \rightarrow SName$.
- $CourseID \rightarrow CourseName, Faculty$.
- $(SID, CourseID) \rightarrow Grade$.

Candidate Key

(SID, CourseID) is the candidate key because a student can enroll in multiple courses and a course can contain multiple students; their combination determines the grade.

Partial Dependencies

SName depends only on SID, and CourseName and Faculty depend only on CourseID. These are partial dependencies on parts of the composite key, so the original relation is not in 2NF.

2NF Decomposition

Relation	Attributes	Primary Key
Student	SID, SName	SID
Course	CourseID, CourseName, Faculty	CourseID
Enrollment	SID, CourseID, Grade	(SID, CourseID)

The partial dependencies are removed. In Enrollment, Grade depends on the complete composite key. Hence the decomposition is in 2NF.

Question 4

Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.

SOL : Functional Dependencies

- $PassID \rightarrow PassName$.

- $\text{FlightNo} \rightarrow \text{DeptCity}, \text{ArrCity}, \text{Date}$.
- $(\text{FlightNo}, \text{PassID}) \rightarrow \text{Seat}$.

Candidate Key

$(\text{FlightNo}, \text{PassID})$ is the candidate key because a passenger can book different flights and a flight can have multiple passengers.

BCNF Violations

PassID and FlightNo are not superkeys of the original Booking relation, yet they determine other attributes. Therefore, the original relation violates BCNF.

BCNF Decomposition

Relation	Attributes	Primary Key
Passenger	$\text{PassID}, \text{PassName}$	PassID
Flight	$\text{FlightNo}, \text{DeptCity}, \text{ArrCity}, \text{Date}$	FlightNo
Booking	$\text{FlightNo}, \text{PassID}, \text{Seat}$	$(\text{FlightNo}, \text{PassID})$

In Passenger, PassID is the key. In Flight, FlightNo is the key. In Booking, $(\text{FlightNo}, \text{PassID})$ is the key. Thus, every determinant in the decomposed relations is a superkey, so the schema is in BCNF.

Question 5

Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.

SOL : Assumption for a standard HR schema: $\text{Employee}(\text{EmpID}, \text{EmpName}, \text{DeptID}, \text{DeptName}, \text{ManagerID}, \text{ManagerName}, \text{CountryID}, \text{CountryName}, \text{Salary})$. Each employee belongs to one department and one country; each department has one manager; IDs uniquely identify their corresponding names.

Functional Dependencies

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}, \text{ManagerID}, \text{CountryID}, \text{Salary}$.
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$.
- $\text{ManagerID} \rightarrow \text{ManagerName}$.
- $\text{CountryID} \rightarrow \text{CountryName}$.

Candidate Key

EmpID is the candidate key for the assumed HR relation.

Transitive Dependencies

- $\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$.
- $\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{ManagerID} \rightarrow \text{ManagerName}$.
- $\text{EmpID} \rightarrow \text{CountryID} \rightarrow \text{CountryName}$.

These are transitive because non-key attributes such as DeptID and CountryID determine other non-key attributes.

3NF Decomposition

Relation	Attributes	Primary Key
Employee	EmpID, EmpName, DeptID, ManagerID, CountryID, Salary	EmpID
Department	DeptID, DeptName, ManagerID	DeptID
Manager	ManagerID, ManagerName	ManagerID
Country	CountryID, CountryName	CountryID

The decomposition removes transitive dependencies. Employee stores employee-specific facts, Department stores department facts, Manager stores manager facts, and Country stores country facts. Each non-key attribute depends directly on the key of its relation, so the design satisfies 3NF.

Question 6

Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.

SOL : Functional Dependencies and Multivalued Dependencies

- $\text{MemberID} \rightarrow \text{MemberName}$.
- $\text{BookID} \rightarrow \text{Title}$.
- $\text{BookID} \twoheadrightarrow \text{Author}$ (a book may have multiple authors).
- $\text{BookID} \twoheadrightarrow \text{Genre}$ (a book may have multiple genres).
- $(\text{MemberID}, \text{BookID}) \rightarrow \text{BorrowDate}$.

The important multivalued dependencies are $\text{BookID} \twoheadrightarrow \text{Author}$ and $\text{BookID} \twoheadrightarrow \text{Genre}$. They represent independent multi-valued facts about a book.

4NF Decomposition

Relation	Attributes	Primary Key
Member	MemberID, MemberName	MemberID
Book	BookID, Title	BookID
BookAuthor	BookID, Author	(BookID, Author)
BookGenre	BookID, Genre	(BookID, Genre)
Borrow	MemberID, BookID, BorrowDate	(MemberID, BookID)

The independent author and genre multivalues are separated. The borrowing relationship is also separated from book details. Hence the decomposition removes the 4NF violation and avoids unnecessary combinations of authors, genres, and borrowing records.

Question 7

Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.

Assumption for analysis: TelecomBilling(CustomerID, CustomerName, PhoneNo, PlanID, PlanName, MonthlyCharge, CallID, CallDate, Duration, CallCharge). CustomerID, PhoneNo, PlanID, and CallID are assumed to be unique identifiers as appropriate. A customer may have multiple calls, and each call belongs to one customer.

SOL : Functional Dependencies

- $\text{CustomerID} \rightarrow \text{CustomerName}, \text{PhoneNo}, \text{PlanID}$.
- $\text{PhoneNo} \rightarrow \text{CustomerID}, \text{CustomerName}, \text{PlanID}$.
- $\text{PlanID} \rightarrow \text{PlanName}, \text{MonthlyCharge}$.
- $\text{CallID} \rightarrow \text{CustomerID}, \text{CallDate}, \text{Duration}, \text{CallCharge}$.
- $(\text{CustomerID}, \text{CallID}) \rightarrow \text{CallDate}, \text{Duration}, \text{CallCharge}$ (redundant if CallID is globally unique).

Candidate Keys

- CustomerID is a candidate key for the customer information portion.
- PhoneNo is also a candidate key if phone numbers are unique.

- CallID is a candidate key for the call/billing portion if each call has a globally unique ID.
- For the full relation, if CallID is globally unique, CallID is the candidate key because it determines the call and its customer; customer details follow through CustomerID.

Primary Key

A suitable primary key for the full billing record is CallID, assuming CallID is unique. CustomerID and PhoneNo are candidate keys for customer records, while PlanID identifies a plan.

Summary Table

Item	Determines
CustomerID	CustomerName, PhoneNo, PlanID
PhoneNo	CustomerID, CustomerName, PlanID
PlanID	PlanName, MonthlyCharge
CallID	CustomerID, CallDate, Duration, CallCharge

Before normalization, the schema contains repeated customer and plan information for every call. This can lead to update, insertion, and deletion anomalies. The identified keys and dependencies provide the basis for decomposition into Customer, Plan, and Call/Billing relations during normalization.

Question 8

A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.

SOL : Given Relation

The original relation is:

Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)

Assumptions and Functional Dependencies

- PID uniquely identifies a product.
- Each SupplierID identifies exactly one supplier name.
- A product can be supplied by one supplier in this simplified schema.
- The category, price, and warehouse information belong to the product identified by PID.

Functional dependencies:

- $PID \rightarrow PName, SupplierID, Category, Price, Warehouse$

- $\text{SupplierID} \rightarrow \text{SupplierName}$

Therefore, $\text{PID} \rightarrow \text{SupplierName}$ also follows transitively through SupplierID .

Candidate Key : Since PID determines all attributes in the relation, PID is the candidate key and is selected as the primary key.

First Normal Form (1NF) : The relation is already in 1NF because every attribute contains atomic values and there are no repeating groups.

$R1 = \text{Product}(\text{PID}, \text{PName}, \text{SupplierID}, \text{SupplierName}, \text{Category}, \text{Price}, \text{Warehouse})$

Second Normal Form (2NF) : The candidate key consists of a single attribute, PID . Therefore, partial dependency is not possible. Hence the relation is already in 2NF.

Third Normal Form (3NF) : The relation is not in 3NF because of the transitive dependency:

$\text{PID} \rightarrow \text{SupplierID} \rightarrow \text{SupplierName}$

SupplierName depends on SupplierID rather than directly on the key PID . To remove this transitive dependency, decompose the relation.

Relation	Attributes	Primary Key	Foreign Key
PRODUCT	PID, PName, SupplierID, Category, Price, Warehouse	PID	$\text{SupplierID} \rightarrow \text{SUPPLIER.SupplierID}$
SUPPLIER	SupplierID, SupplierName	SupplierID	—

BCNF Verification

- In **PRODUCT**, PID is the determinant of all product attributes and PID is a candidate key. Therefore **PRODUCT** satisfies BCNF.
- In **SUPPLIER**, SupplierID determines SupplierName and SupplierID is the candidate key. Therefore **SUPPLIER** satisfies BCNF.

Final Normalized Design

The complete normalized design is:

$\text{PRODUCT}(\text{PID}, \text{PName}, \text{SupplierID}, \text{Category}, \text{Price}, \text{Warehouse})$

SUPPLIER(SupplierID, SupplierName)

This design eliminates the transitive dependency and reduces update, insertion, and deletion anomalies.

Question 9

Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.

SOL : Poorly Normalized School ERP Schema

Since the question does not provide the exact attributes, the following representative School ERP relation is assumed:

SchoolERP(StudentID, StudentName, DeptID, DeptName, CourseID, CourseName, TeacherID, TeacherName, TeacherPhone, Semester, Grade)

Functional Dependencies

The following functional dependencies are assumed for a typical School/College ERP:

- StudentID $\rightarrow$ StudentName, DeptID
- DeptID $\rightarrow$ DeptName
- CourseID $\rightarrow$ CourseName
- TeacherID $\rightarrow$ TeacherName, TeacherPhone
- StudentID, CourseID, Semester $\rightarrow$ TeacherID, Grade

The composite key of the original enrollment relation is assumed to be (StudentID, CourseID, Semester), because a student may take many courses and may have different enrollments across semesters.

Problems in the Original Relation

- Student details are repeated for every course taken by the same student.
- Department information is repeated for every student in the department.
- Course information is repeated for every enrollment.
- Teacher information is repeated for every course enrollment.
- Updating a department, course, or teacher may require many rows to be changed.
- Insertion and deletion anomalies can occur because unrelated information is stored together.

First Normal Form (1NF)

The relation is converted to 1NF by ensuring that all attributes contain atomic values and that each row represents one student-course-semester enrollment.

Second Normal Form (2NF)

The key is (StudentID, CourseID, Semester). Several non-key attributes depend only on part of the composite key:

- StudentID $\rightarrow$ StudentName, DeptID
- CourseID $\rightarrow$ CourseName
- TeacherID $\rightarrow$ TeacherName, TeacherPhone

These partial dependencies are removed by decomposition.

Relation	Attributes	Primary Key
STUDENT	StudentID, StudentName, DeptID	StudentID
DEPARTMENT	DeptID, DeptName	DeptID
COURSE	CourseID, CourseName	CourseID
TEACHER	TeacherID, TeacherName, TeacherPhone	TeacherID
ENROLLMENT	StudentID, CourseID, Semester, TeacherID, Grade	StudentID + CourseID + Semester

Third Normal Form (3NF)

The transitive dependency DeptID $\rightarrow$ DeptName is removed by creating the DEPARTMENT relation. Similarly, TeacherID $\rightarrow$ TeacherName, TeacherPhone is stored in the TEACHER relation. Thus no non-key attribute depends transitively on a key.

BCNF Decomposition

The resulting relations satisfy BCNF under the stated assumptions because every non-trivial determinant is a candidate key in its relation:

- STUDENT: StudentID is the determinant and candidate key.
- DEPARTMENT: DeptID is the determinant and candidate key.

- COURSE: CourseID is the determinant and candidate key.
- TEACHER: TeacherID is the determinant and candidate key.
- ENROLLMENT: (StudentID, CourseID, Semester) is the candidate key under the stated assumptions.

Final BCNF Design

STUDENT(StudentID, StudentName, DeptID)

DEPARTMENT(DepID, DeptName)

COURSE(CourseID, CourseName)

TEACHER(TeacherID, TeacherName, TeacherPhone)

ENROLLMENT(StudentID, CourseID, Semester, TeacherID, Grade)

Question 10

Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.

SOL : Assumed Movie Streaming Schema

Because the question does not specify the exact attributes, consider the following relation:

STREAMING(MovieID, MovieTitle, ActorID, ActorName, LanguageID, LanguageName, PlatformID, PlatformName)

Key Assumption

Assume that a movie may have multiple actors, multiple languages, and may be available on multiple streaming platforms. The independent many-to-many relationships are:

- Movie $\leftrightarrow$ Actor
- Movie $\leftrightarrow$ Language
- Movie $\leftrightarrow$ Platform

The natural key of the original relationship is the combination (MovieID, ActorID, LanguageID, PlatformID) if a row represents one valid combination of these independent associations.

Join Dependencies

The important join dependency is:

$*(\text{MovieID}, \text{ActorID}, \text{LanguageID}, \text{PlatformID}) = *(\text{MovieID}, \text{ActorID}, \text{LanguageID}) \quad (\text{MovieID}, \text{ActorID}, \text{PlatformID}) \quad (\text{MovieID}, \text{LanguageID}, \text{PlatformID})$

This means that the original relation can be represented as the lossless join of smaller relations when the three relationships are independent.

5NF Decomposition

Decompose the relation into the following relations:

Relation	Attributes	Primary Key
MOVIE_ACTOR	MovieID, ActorID	MovieID + ActorID
MOVIE_LANGUAGE	MovieID, LanguageID	MovieID + LanguageID
MOVIE_PLATFORM	MovieID, PlatformID	MovieID + PlatformID

Lookup Relations

Descriptive details are stored separately to avoid redundancy:

Relation	Attributes	Primary Key
MOVIE	MovieID, MovieTitle	MovieID
ACTOR	ActorID, ActorName	ActorID
LANGUAGE	LanguageID, LanguageName	LanguageID
PLATFORM	PlatformID, PlatformName	PlatformID

5NF Verification

A relation is in 5NF (Project-Join Normal Form) when every non-trivial join dependency is implied by the candidate keys. After decomposition, the independent many-to-many relationships are represented separately:

- MOVIE_ACTOR contains only the Movie–Actor relationship.
- MOVIE_LANGUAGE contains only the Movie–Language relationship.
- MOVIE_PLATFORM contains only the Movie–Platform relationship.
- MOVIE, ACTOR, LANGUAGE, and PLATFORM contain descriptive information identified by their respective keys.

Therefore, under the stated independence assumption, the design satisfies 5NF.

Lossless-Join Verification

The original relationship can be reconstructed by joining the decomposed relationship tables on their common MovieID:

R = MOVIE_ACTOR MOVIE_LANGUAGE MOVIE_PLATFORM

The joins are lossless when the business rule guarantees that the three associations are independent and that only valid combinations represented by the decomposed relations are intended.

Example

Suppose Movie M1 has actors A1 and A2, languages L1 and L2, and is available on platforms P1 and P2. The decomposed tables store these relationships independently:

- MOVIE_ACTOR: (M1,A1), (M1,A2)
- MOVIE_LANGUAGE: (M1,L1), (M1,L2)
- MOVIE_PLATFORM: (M1,P1), (M1,P2)

The original relation can be reconstructed through joins. No actor, language, or platform name needs to be repeated in every combined row.

Final Answer:

The final 5NF design is:

MOVIE(MovieID, MovieTitle)

ACTOR(ActorID, ActorName)

LANGUAGE(LanguageID, LanguageName)

PLATFORM(PlatformID, PlatformName)

MOVIE_ACTOR(MovieID, ActorID)

MOVIE_LANGUAGE(MovieID, LanguageID)

MOVIE_PLATFORM(MovieID, PlatformID)

This decomposition removes redundancy caused by the join dependency and provides a lossless representation under the stated business rules.